Resampling-Based Model Selection for PE SDFs

How to empirically measure the risk-adjusted performance of non-traded cash flows

April 22, 2026

1 Idea

The paper asks a simple empirical question: which stochastic discount factor (SDF) models give the most credible risk adjustment for private-equity cash flows?

Three difficulties make this question hard to answer with existing methods.

Difficulty 1: moment construction matters. Private-equity samples are small, lumpy, and organized by vintage year. An SDF model can look good or bad depending on how funds are grouped into pricing moments. Existing approaches commit to one grouping—typically single funds or vintage-year portfolios—and treat it as fixed. But the choice of grouping is itself consequential for estimation, and there is no a priori reason to prefer one admissible grouping over another.

Difficulty 2: unseasoned vintages create a dilemma. Recent vintage years carry valuable pricing information—they reflect current market conditions and expand the factor-path coverage of the sample—but including them naively destabilizes estimation. The standard approach treats the latest reported NAV as a terminal cash flow and discounts it over the fund's entire remaining horizon. This creates two problems. First, it concentrates a large share of the fund's discounted value at a single calendar date—the dataset endpoint—so that the estimator's fit becomes disproportionately sensitive to the macro state at that date rather than to the fund's lifetime factor exposure. Second, the terminal NAV is an appraisal, not a realized payoff; its co-movement with public indices may differ systematically from that of actual divestitures. Researchers therefore face a dilemma: exclude recent vintages and waste data, or include them and accept fragile estimates driven by one macro-state realization.

Difficulty 3: exploding-alpha degeneracy in finite samples. Independently of the unseasoned-vintage problem, NPV-style SDF estimation suffers from a finite-sample pathology first documented by Driessen et al. [2012]. In the standard multiplicative SDF parameterization, cash flows are discounted by a cumulative factor that includes $(1+\alpha)^{-T}$. When α is large and positive, this factor drives all discount factors toward zero for large T, making every fund's pricing error approximately zero regardless of the factor loadings β. The result is a ridge of spurious near-minima along the alpha axis of the objective function that prevents reliable joint identification of α and β. The problem is most severe when fund horizons are long and early cash flows are small relative to later ones, which is precisely the cash-flow shape of a typical PE fund. Existing mitigants include ratio or log transformations of the pricing restriction [Driessen et al., 2008], horizon averaging, and box constraints on α, but none eliminates the underlying mechanism.

We propose a *NAV-bridged cash-flow estimator* that addresses all three difficulties. The estimator has two components:

1. **Hybrid pricing restrictions.** For seasoned vintages whose funds are substantially liquidated, the estimator uses the standard cumulative cash-flow pricing error. For unseasoned vintages with significant residual NAVs, it replaces the single long-horizon pricing restriction with a sequence of one-period Euler restrictions on quarterly NAV returns. Each quarterly Euler error involves only one-period discounting, so its sensitivity to alpha is proportional to the quarter length rather than exponential in the fund horizon. The key theoretical property is that the discounted Euler errors *telescope*: for a fully liquidated fund, the sum of discounted quarterly Euler errors equals the standard cumulative cash-flow error exactly.

The estimator is therefore asymptotically a pure cash-flow estimator; the Euler NAV restrictions are finite-sample corrections that vanish as funds mature and the vintage panel grows.

2. **Resampling-based model selection.** Instead of committing to one grouping of funds into pricing moments, we make the grouping itself part of the empirical design. We repeatedly draw random within-vintage fund partitions, estimate the SDF on partition-induced moments, and evaluate pricing errors on held-out moments. The preferred SDF model is the one with the lowest average holdout pricing loss across many admissible partitions.

The method therefore has three goals:

- **Efficient use of the vintage panel:** include all available vintages—seasoned and unseasoned—without triggering exploding-alpha instability.
- **Model selection:** rank SDF models by out-of-sample pricing performance across many resampled moment constructions.
- **Design comparison:** compare whether within-vintage fund resampling or traditional vintage-year portfolio construction yields more stable and predictive pricing moments.

The paper is closest in spirit to Korteweg and Nagel [2016] because the estimator works with aggregated pricing moments, and to Driessen et al. [2012] because the moments price private cash flows directly. The new contributions are the hybrid CF/Euler pricing restriction that nests the pure cash-flow estimator asymptotically, and the resampling layer that treats moment construction as a testable design choice rather than a fixed input. Brown et al. [2023] also use NAVs to estimate the risk and return of PE funds.

2 Guiding Example

Suppose a candidate SDF has three parameters: an alpha term and two factor loadings. Then we need at least three pricing moments to estimate it. A natural resampling design is therefore:

- build three estimation folds,
- build one validation fold,
- estimate the SDF on the three estimation moments,
- score the fitted SDF on the validation moment.

Figures 1 and 2 show the two benchmark ways to build these folds. In both figures, each dot is one fund and the active vintages are 2005–2010. Vintages 2011–2012 are excluded because they are too young or not sufficiently liquidated.

3 Two Canonical Sampling Schemes

Proposed split: within-vintage fund resampling. In the proposed design, every fold contains funds from every active vintage. This keeps the vintage-year support fixed across estimation and validation moments. The folds differ because they contain different funds, not because they represent different macro or vintage states. This is the cleanest design for asking whether an SDF prices PE cash flows robustly across fund-level resamples.

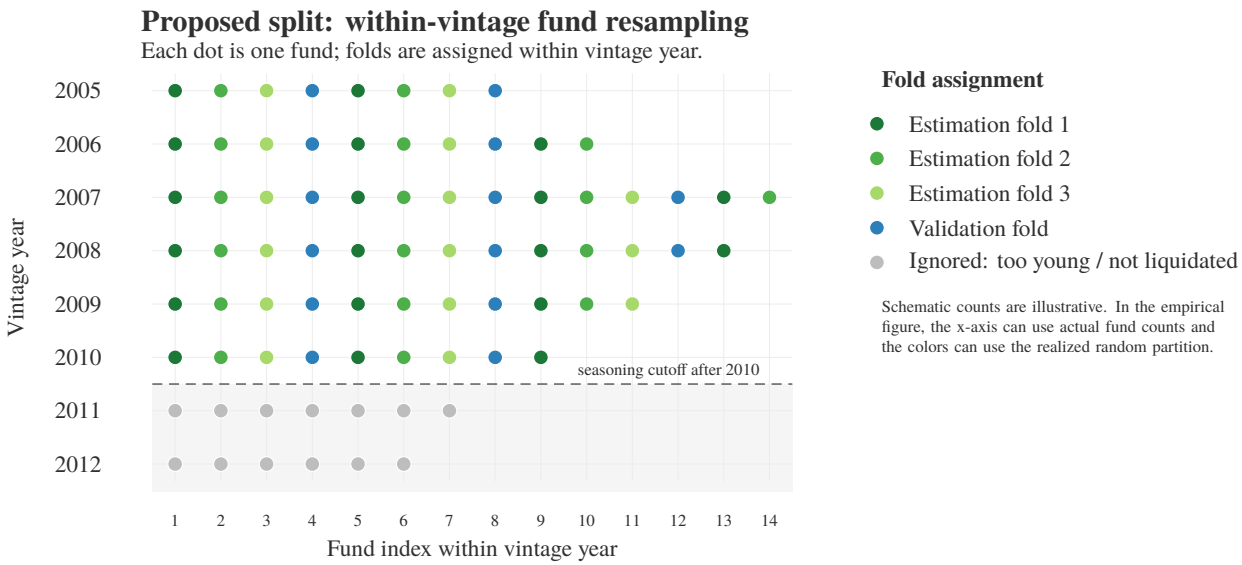

Figure 1: Proposed within-vintage fund resampling scheme. Each active vintage is split into three estimation cells and one validation cell. Fold 1 contains the dark-green funds from every vintage, fold 2 contains the medium-green funds from every vintage, fold 3 contains the light-green funds from every vintage, and the validation fold contains the blue funds from every vintage.

Benchmark split: shuffled vintage-year portfolios. In the traditional design, the primitive object is a vintage-year portfolio. A fold contains whole vintages, not fund sub-samples from all vintages. This is easy to explain and resembles standard PE portfolio construction. But the validation moment now differs from the estimation moments in vintage composition. A failure on the validation fold may therefore reflect poor SDF pricing, or it may reflect extrapolation across vintage states.

The first version of the paper should focus on this contrast. If within-vintage resampling delivers more stable estimates and lower holdout pricing errors, the evidence favors building all-vintage subportfolios before estimating the SDF. If shuffled vintage-year portfolios perform similarly, the simpler vintage-year construction may be sufficient.

4 Objects and Notation

Let $v = 1, \ldots, V$ index active vintage years, and let $\mathcal{I}_v$ be the set of funds in vintage v. Let M denote a candidate SDF model with parameter vector $\theta \in \Theta_M \subset \mathbb{R}^{p(M)}$.

We use P estimation folds and h holdout folds. For the guiding three-parameter example, $P = 3$ and $h = 1$. When comparing models with different numbers of parameters, a clean implementation is to use a common number of estimation folds $P = p_{\max}$, where $p_{\max} = \max_M p(M)$.

4.1 Within-vintage resampling

For sampled split $n = 1, \ldots, N$, split the funds within each vintage into $P + h$ cells:

$$\mathcal{P}_{n,v} = \{C_{n,v,1}, \ldots, C_{n,v,P+h}\}, \qquad C_{n,v,r} \subset \mathcal{I}_v.$$

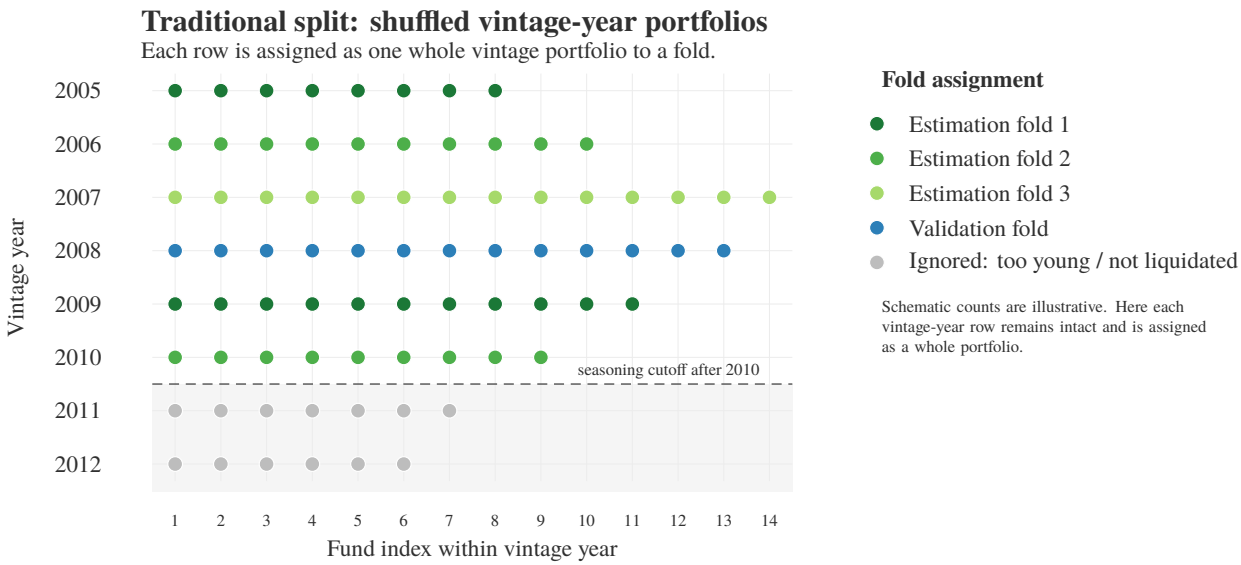

Figure 2: Traditional shuffled vintage-year portfolio scheme. Funds are first aggregated into whole vintage-year portfolios, and whole vintages are then assigned to folds. Here, 2005 and 2009 enter estimation fold 1, 2006 and 2010 enter estimation fold 2, 2007 enters estimation fold 3, and 2008 is the validation fold.

The cells are disjoint and cover the funds in vintage v. The rth cross-vintage fold is

$$F_{n,r} = \{C_{n,1,r}, C_{n,2,r}, \ldots, C_{n,V,r}\}.$$

Thus fold r contains one fund sub-sample from every active vintage.

At the single-fund level, the number of admissible assignments is usually too large to enumerate. We therefore draw Monte Carlo samples from a partition rule Π:

$$\mathcal{P}_n \sim \Pi(\cdot \mid P, h, \mathcal{I}_1, \ldots, \mathcal{I}_V).$$

For now, Π should be simple and transparent: random assignment within each vintage, subject to no empty cells and reasonable balance in fund counts or economic size.

4.2 Vintage-year portfolio benchmark

For the benchmark design, first aggregate all funds in each vintage v into a vintage-year portfolio. Then assign whole vintages to folds. The notation is the same at the fold level, but the economic object differs: $F_{n,r}$ now contains whole vintage-year portfolios rather than within-vintage fund cells.

5 Pricing Moments

For a fold $F_{n,r}$, let $\epsilon_{n,v,r}(\theta; M, \eta)$ be the pricing error from the vintage component of that fold. The design choice η collects choices such as the alpha definition, loss function, and cash-flow weighting.

The aggregated fold moment is

$$m_{n,r}(\theta; M, \eta) = \sum_{v \in A_{n,r}} \omega_{n,v,r}\, \epsilon_{n,v,r}(\theta; M, \eta),$$

where $A_{n,r}$ is the set of vintages represented in fold r, and $\omega_{n,v,r}$ are vintage or value weights. Under within-vintage resampling, $A_{n,r}$ is the full active vintage set for every fold. Under shuffled vintage-year portfolios, $A_{n,r}$ is only the set of whole vintages assigned to fold r.

This distinction is the core empirical design choice. The proposed method creates comparable fold moments that all span the same vintage domain. The benchmark creates moments from different vintage domains.

5.1 Fund weights and vintage weights

The fund-level weight w_i and vintage weight $\omega_{n,v,r}$ are design choices that determine the economic content of the fold moment. The baseline specification sets $w_i = 1/K_i$, where K_i is committed capital, so that each fund contributes a *relative* pricing error (per dollar committed). At the cell level, we average over the funds in the cell. At the vintage level, the baseline sets $\omega_{n,v,r} = 1/|A_{n,r}|$ (equal vintage weights), so that each vintage contributes equally to the fold moment regardless of the number of funds or aggregate capital it contains.

Under this three-layer normalization (fund → cell → vintage), the fold moment becomes

$$m_{n,r}(\theta; M, \eta) = \frac{1}{|A_{n,r}|} \sum_{v \in A_{n,r}} \frac{1}{|C_{n,v,r}|} \sum_{i \in C_{n,v,r}} \frac{\epsilon_i(\theta; M, \eta)}{K_i}.$$

This is an average, across vintages, of the average relative pricing error per fund. It has a clean economic interpretation (mean risk-adjusted NPV per dollar per fund per vintage) and is comparable across: (i) vintages with different numbers of funds, (ii) vintages with different aggregate capital, (iii) seasoned vintages with cash-flow errors and unseasoned vintages with Euler errors (Section 9), and (iv) estimation folds and validation folds.

6 Estimation and Holdout Scoring

For each sampled split n, let E_n be the set of estimation folds and H_n the validation fold. The split-specific estimator solves

$$\hat{\theta}_n(M, \eta, \Pi) = \arg\min_{\theta \in \Theta_M} \frac{1}{P} \sum_{j \in E_n} L\big(m_{n,j}(\theta; M, \eta)\big).$$

The validation score is the pricing loss on the held-out fold:

$$S_n(M, \eta, \Pi) = L\Big(m_{n,H_n}(\hat{\theta}_n(M, \eta, \Pi); M, \eta)\Big).$$

Aggregate this score across many sampled assignments:

$$\text{Score}_N(M, \eta, \Pi) = \frac{1}{N} \sum_{n=1}^{N} S_n(M, \eta, \Pi).$$

The preferred model-design pair (M, η, Π) is the one with the lowest average holdout pricing loss. Parameter stability is still informative, but the primary ranking criterion should be predictive pricing performance: does the fitted SDF price cash flows that were not used in estimation?

6.1 Choice of Pricing Moment

The pricing error $m_{n,j}(\theta; M, \eta)$ in the loss L is a design choice, not a primitive. Two canonical families appear in the private-capital SDF literature: the *dollar NPV* moment of Driessen et al. [2012] and the *PME-ratio* moment of Kaplan and Schoar [2005], Driessen et al. [2009], and Ang et al. [2018]. Because the resampling-based ranking of (M, η, Π) can in principle be sensitive to this choice, we discuss both here and treat the NPV moment as the baseline, the PME-ratio as a robustness check.

Dollar NPV moment. For fold cell j in split n, aggregate discounted cash flows into one cumulative pricing error,

$$m_{n,j}^{\mathrm{NPV}}(\theta; M, \eta) = \sum_{i \in C_{n,j}} w_i \sum_{t=0}^{T_i} \Psi_{0,t}(\theta; M, \eta)\, CF_{t,i}, \qquad CF_{t,i} = D_{t,i} - X_{t,i},$$

where $X_{t,i}$ are contributions, $D_{t,i}$ distributions (including terminal NAV for seasoned funds), and $\Psi_{0,t}$ the cumulative SDF. At the population parameter, $E[m_{n,j}^{\mathrm{NPV}}(\theta_0)] = 0$: the risk-adjusted NPV of fund cash flows vanishes.

Pros. (i) *Linearity* in cash flows yields tractable GMM derivatives, influence functions, and CLT-based inference. (ii) *Additivity* across funds makes the fold structure of Section 11, as well as subsampling and the bootstrap, transparent: the cell moment is the sum of fund-level moments, and the split moment is the sum of cell moments. (iii) *Compatibility with the Euler NAV restriction* for unseasoned vintages: the one-period quarterly Euler error $\eta_{t,i}(\theta)$, discounted to time zero and summed, telescopes into m^{NPV} when $\mathrm{NAV}_{0,i} = \mathrm{NAV}_{T_i,i} = 0$. The same pricing object is therefore used for seasoned and unseasoned vintages, up to a finite-sample NAV correction that vanishes as funds mature. No denominator can be near zero; there is no ratio instability.

Cons. (i) *Scale-dependence.* A \$1B fund with a 1% relative NPV error contributes ten times more to the moment than a \$100M fund with the same relative error, so efficient GMM weighting implicitly tilts the estimator toward mega-funds. The influence function is mechanically heavy-tailed in the fund-size distribution rather than in genuine pricing disagreement. (ii) *Dollar-denominated comparisons across vintages* with heterogeneous aggregate commitments are not directly meaningful; normalization is left to the econometrician. (iii) *Sign cancellation.* A cross-sectional mean of zero is consistent with systematic under-pricing of one subgroup offsetting over-pricing of another, which the scalar moment cannot detect. This motivates the fine-grained fold structure but does not eliminate the concern within a fold.

PME-ratio moment. Define the risk-adjusted PME of fund i as

$$\mathrm{PME}_i(\theta; M, \eta) = \frac{\sum_t \Psi_{0,t}(\theta; M, \eta)\, D_{t,i}}{\sum_t \Psi_{0,t}(\theta; M, \eta)\, X_{t,i}},$$

from which two natural cell-level moments follow: the *average-PME* form $m_{n,j}^{\mathrm{PME}}(\theta) = |C_{n,j}|^{-1} \sum_{i \in C_{n,j}} \big(\mathrm{PME}_i(\theta) - 1\big)$ and the *pooled-PME* form $m_{n,j}^{\text{PME-pool}}(\theta) = \big(\sum_{i,t} \Psi_{0,t} D_{t,i}\big) / \big(\sum_{i,t} \Psi_{0,t} X_{t,i}\big) - 1$. Both have mean zero at θ_0 under standard regularity, but they estimate different population objects (mean of a ratio versus ratio of means).

Pros. (i) *Scale-invariance.* The ratio is unit-free, so fund-size heterogeneity does not mechanically up-weight mega-funds. A \$100M fund and a \$1B fund enter the cross-sectional moment on equal footing. (ii) *Practitioner interpretability.* PME is the native performance metric of the industry; reporting results in

the same units aids dissemination. (iii) *Bounded downside.* The fund-level PME is bounded below by zero, so $\mathrm{PME}_i - 1$ is bounded below by -1: a single disaster fund cannot dominate the moment the way it can dominate a dollar NPV.

Cons. (i) *Nonlinearity.* The moment is a ratio in θ; its derivative involves the ratio of two correlated sums and is numerically fragile when the denominator (discounted contributions) is small. This occurs mechanically for young funds with back-loaded capital calls — precisely the vintages that the Euler NAV restriction is designed to retain. (ii) *Jensen-style bias between forms.* The average-PME and pooled-PME moments can differ in finite samples by a term of order $\mathrm{Var}(X_i)/\bar{X}^2$; the two are asymptotically equivalent only under degenerate size distributions. Researchers must commit to one and defend it. (iii) *No telescoping with the Euler NAV restriction.* Integrating unseasoned vintages $\mathcal{V}^U$ into a PME framework typically requires treating the latest $\mathrm{NAV}_{T_i,i}$ as a terminal distribution and discounting it over the fund's entire remaining horizon — exactly the long-horizon discounting mechanism that motivates ϵ^U in Section 9. The PME-ratio framework therefore reintroduces the exploding-alpha sensitivity for $\mathcal{V}^U$ that the NPV/Euler combination was designed to eliminate. (iv) *Inference.* Upper-tail skewness in PME_i (funds can return many times their capital, but cannot lose more than it) distorts CLT-based standard errors and t-statistics unless explicitly robustified, e.g., by bootstrapping the cell-level moment or by employing median-based loss functions.

Choice for the empirical implementation. The telescoping compatibility with the Euler NAV restriction and the tractability of inference make the dollar NPV moment our primary analytical framework. However, cross-vintage comparability requires normalization: a \$1B fund with a 1% relative NPV error contributes ten times more to the raw dollar moment than a \$100M fund with the same relative error, and seasoned vintages with long cash-flow histories mechanically produce larger-magnitude pricing errors than unseasoned vintages with only a few quarters of Euler errors.

Therefore, the *scale-normalized NPV*, $m_{n,j}^{\mathrm{NPV}} / \sum_{i \in C_{n,j}} K_i$ where K_i is committed capital, is the operational baseline: it preserves linearity, neutralizes the large-fund tilt, and delivers comparable magnitudes across seasoned and unseasoned vintages. This is consistent with the three-layer normalization described in Section 5.1. We additionally report two variants as robustness checks: (a) the raw dollar NPV, to verify that normalization does not materially change model rankings; and (b) the pooled-PME moment restricted to seasoned vintages $\mathcal{V}^S$, which is where PME is unambiguously well-defined. Stability of the preferred (M, η, Π) across these three moments is the appropriate robustness diagnostic: agreement suggests genuine model fit, while disagreement would flag moment-induced identification.

7 Why Validation Is Needed

If a model has p parameters and we estimate it on exactly p moments, the in-sample fit can look deceptively good. This is especially dangerous in small PE samples because exact identification can hide instability. The validation fold creates a more demanding question: after fitting the model on P moments, does it also price a held-out PE cash-flow portfolio?

The main empirical specifications should be:

- **Main comparison:** $P = p_{\max}$, $h = 1$, so all candidate models are compared on a common fold architecture.
- **Exact-identified diagnostic:** $P = p(M)$, $h = 0$, to study feasibility and parameter instability.
- **Model-specific validation:** $P = p(M)$, $h = 1$, to compare exact-identified estimation plus one holdout

fold.

- **Scheme comparison:** run the same model list under within-vintage resampling and shuffled vintage-year portfolios.

8 Alpha Design

The resampling method does not by itself decide what alpha should mean. That choice is central to the paper. The cleanest baseline is to use a cumulative, NPV-style alpha rather than a period-by-period intercept alpha.

8.1 Baseline: cumulative alpha

For fold $F_{n,j}$, let $B_t(\beta)$ be the cumulative benchmark return implied by factor loadings β. A direct-alpha-style fold moment is

$$\psi_{n,j}(a,\beta) = \sum_t \frac{cf_{n,j,t}}{B_t(\beta)} \exp\big(-a\,\tau_{n,j,t}\big),$$

where $cf_{n,j,t}$ is the fold cash flow at time t, and $\tau_{n,j,t}$ is elapsed time from the fold origin to date t. For fixed β, the fold-implied cumulative alpha solves

$$\sum_t \frac{cf_{n,j,t}}{B_t(\beta)} \exp\big(-a_{n,j}(\beta)\,\tau_{n,j,t}\big) = 0.$$

This is close to the KN16/GPME/direct-alpha tradition because alpha is an NPV-style abnormal-performance object [Korteweg and Nagel, 2016, Gredil et al., 2023, Korteweg and Nagel, 2025]. It is also more natural for long-horizon PE cash flows than a monthly intercept alpha.

The fold-implied alpha $a_{n,j}(\beta)$ serves a dual purpose in the resampling design. In *estimation*, a common alpha a is estimated jointly with β across the P estimation folds by minimizing the aggregate loss over fold moments $\psi_{n,j}(a,\beta)$. In *diagnostics*, the fold-specific implied alpha $a_{n,j}(\hat{\beta})$ is computed for each fold (including the validation fold) by solving $\psi_{n,j}(a,\hat{\beta}) = 0$ at the estimated $\hat{\beta}$. The cross-fold dispersion of $\{a_{n,j}\}_{j=1}^{P+h}$ is then a model diagnostic: a large spread indicates that the alpha estimate is sensitive to which funds enter the fold, and therefore that the aggregate alpha is weakly identified. This parallels how KN16 treat GPME: one SDF parameter vector is estimated; the implied alpha per asset class is a derived diagnostic quantity.

Compounding convention. The exponential $\exp(-a\,\tau)$ parameterizes alpha as a continuously compounded rate, in contrast to the discretely compounded $(1+\alpha)^{-t}$ used in the main SDF paper. For estimation the two are equivalent up to the reparameterization $a = \log(1+\alpha)$. The continuous form yields cleaner analytical derivatives and avoids the product notation of the discrete SDF, which simplifies the optimization. All reported alpha estimates can be converted back to the discrete convention via $\alpha = e^a - 1$.

8.2 Alternative alpha definitions

The paper should still compare alpha definitions:

- **Cumulative alpha:** the recommended baseline, estimated from benchmark-discounted cash flows.

- **Error-term alpha:** alpha as the mean pricing-error component, as in the machine-learning approach of Tausch and Pietz [2024].
- **Intercept alpha:** a DLP12-style robustness specification, useful for comparison but vulnerable to exploding-alpha behavior in small or weakly identified samples [Driessen et al., 2012, 2008].

The empirical question is not only which SDF factors perform best, but also which alpha definition yields stable and predictive PE pricing moments.

8.3 Asymptotic irrelevance of the exploding-alpha degeneracy

The exploding-alpha issue—where an arbitrarily large α drives cumulative discount factors to zero, creating a ridge of spurious near-minima—is a *finite-sample* pathology, not an asymptotic one. Under increasing-domain asymptotics indexed by vintage years, as $V \to \infty$ the cross-vintage moment averages over many independent realizations of the cumulative SDF. The global minimum of the population criterion $Q(\theta)$ is uniquely attained at θ_0 under standard identification conditions, and the spurious near-minima created by extreme alpha values vanish: no single alpha value can simultaneously zero out the discounted cash flows of many independent vintage realizations with heterogeneous timing and factor paths.

This asymptotic argument justifies the cumulative cash-flow estimator as the target. In finite samples, the exploding-alpha ridge is controlled by two complementary mechanisms:

1. **Euler NAV restrictions.** For unseasoned vintages, the one-period Euler restrictions of Section 9 replace long-horizon cumulative discounting with quarterly discounting. The sensitivity of $\eta_{t,i}$ to α is proportional to the quarter length Δt, not exponential in the fund horizon T. This removes the most vulnerable vintages—those with large residual NAVs and long remaining horizons—from the cumulative-discounting mechanism that generates the ridge.
2. **Resampling regularization.** The validation fold penalizes degenerate alpha values because extreme parameter estimates that overfit the estimation folds will produce large holdout pricing errors. Across many resampled splits, degenerate solutions are revealed by high holdout-loss variance and frequent boundary solutions.

Together, these mechanisms ensure that the exploding-alpha issue is a finite-sample nuisance controlled by the estimation design, not a fundamental limitation of the SDF pricing framework. The estimator is therefore asymptotically a pure cash-flow estimator; the Euler NAV corrections and alpha constraints are finite-sample stabilization devices that vanish in the limit.

9 Euler NAV Restrictions for Unseasoned Vintages

The empirical algorithm in Section 11 excludes insufficiently seasoned vintages. This wastes data from recent vintages that carry useful pricing information. Including them naively—by treating the latest reported NAV as a terminal distribution—amplifies alpha instability, because a single large NAV position is discounted over the fund's entire remaining horizon.

We propose a different treatment. For unseasoned vintages, replace the single cumulative pricing restriction with a sequence of one-period Euler restrictions on quarterly NAV returns. This retains the information from recent vintages while avoiding long-horizon discounting. The resulting estimator is a *NAV-bridged cash-flow*

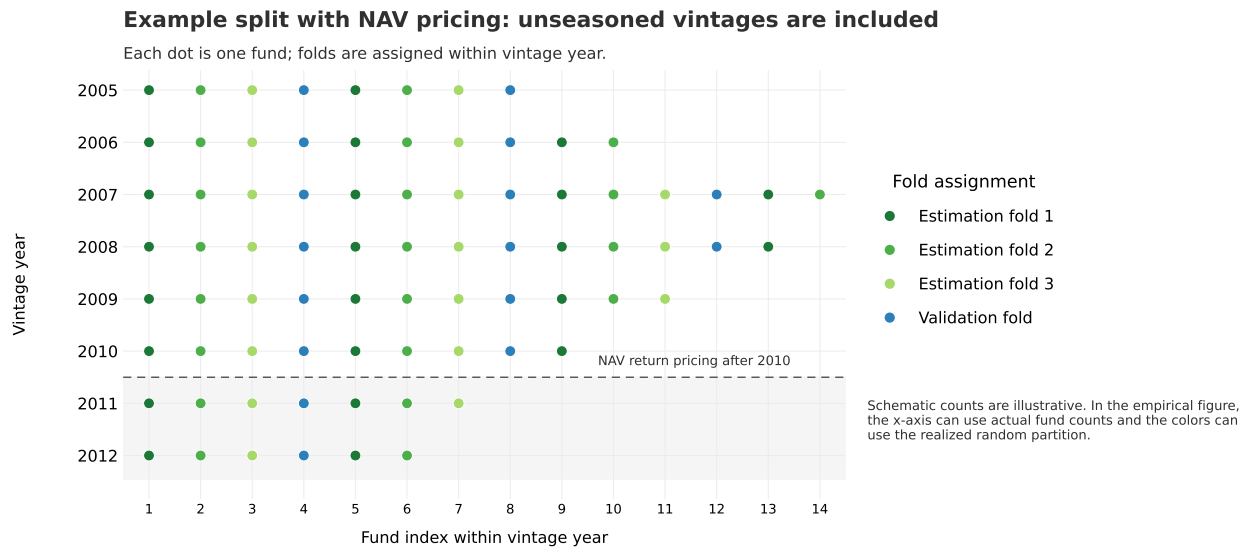

Figure 3: Within-vintage fold assignment under the NAV-bridged cash-flow estimator. Unseasoned vintages 2011–2012 ($\mathcal{V}^U$, grey band) are included in estimation and validation like the seasoned vintages ($\mathcal{V}^S$), but contribute the one-period Euler NAV error ϵ^U in place of the cumulative cash-flow error ϵ^S.

estimator: asymptotically a pure cash-flow estimator, with well-conditioned finite-sample corrections for young funds.

9.1 Seasoned and unseasoned vintage sets

Partition the active vintages into seasoned and unseasoned sets,

$$\mathcal{V} = \mathcal{V}^S \cup \mathcal{V}^U.$$

A vintage v is *unseasoned* ($v \in \mathcal{V}^U$) if its aggregate residual NAV exceeds a fraction δ of total committed capital; otherwise it is *seasoned* ($v \in \mathcal{V}^S$). In practice, $\delta \in [0.10, 0.20]$ is a sensible range. For seasoned vintages, the pricing restriction is the standard cumulative cash-flow error as defined in Section 5. For unseasoned vintages, we use the one-period Euler pricing error defined next.

9.2 One-period Euler pricing error

For a fund i in an unseasoned vintage with quarterly NAV observations at dates $t = 0, 1, \ldots, T_i$, define the one-period pricing error for quarter t:

$$\eta_{t,i}(\theta) = \Psi_{t-1,t}(\theta) \cdot \left(CF_{t,i} + \mathrm{NAV}_{t,i}\right) - \mathrm{NAV}_{t-1,i},$$

where $CF_{t,i}$ is the net cash flow during the quarter, $\mathrm{NAV}_{t,i}$ is the reported net asset value at quarter-end, and $\Psi_{t-1,t}(\theta)$ is the one-period stochastic discount factor.

The restriction $E[\eta_{t,i}(\theta_0)] = 0$ is an Euler equation: the beginning-of-period NAV equals the risk-adjusted present value of within-period distributions plus the end-of-period NAV. It is the same object that prices a one-period return on a traded asset, applied here to the NAV-implied gross return of the fund.

For an unseasoned vintage $v \in \mathcal{V}^U$ and fund cell $C_{n,v,r}$, define the cell-level Euler pricing error as the discounted sum of quarterly errors:

$$\epsilon^U_{n,v,r}(\theta) = \sum_{i \in C_{n,v,r}} w_i \sum_{t=1}^{T_v} \Psi_{0,t-1}(\theta)\, \eta_{t,i}(\theta).$$

The cumulative discount factor $\Psi_{0,t-1}(\theta)$ converts each quarterly error to time-zero units, ensuring that ϵ^U is denominated in the same present-value units as the standard cash-flow pricing error ϵ^S for seasoned vintages. However, unit compatibility does not imply magnitude compatibility: a seasoned vintage contributing 120 quarters of discounted cash flows will mechanically produce larger-magnitude pricing errors than a 2-year-old unseasoned vintage contributing 8 quarters of Euler errors, even if both are correctly priced. The three-layer normalization of Section 5.1—dividing by committed capital at the fund level and averaging across funds and vintages—mitigates this scale gap but does not fully eliminate the variance difference between ϵ^S and ϵ^U.

To diagnose whether the estimator disproportionately fits one group, the empirical implementation should report separate holdout losses for $\mathcal{V}^S$ and $\mathcal{V}^U$:

$$S^S_n(M, \eta, \Pi) = L\left(\sum_{v \in \mathcal{V}^S \cap H_n} \omega_{n,v,H_n}\, \epsilon^S_{n,v,H_n}(\hat{\theta}_n) \right), \qquad S^U_n(M, \eta, \Pi) = L\left(\sum_{v \in \mathcal{V}^U \cap H_n} \omega_{n,v,H_n}\, \epsilon^U_{n,v,H_n}(\hat{\theta}_n) \right).$$

If the average $\bar{S}^U$ is systematically lower than $\bar{S}^S$, the Euler restrictions are being trivially satisfied and contribute little to identification; if $\bar{S}^U \gg \bar{S}^S$, the short NAV histories are dominating the holdout loss and the model selection may be driven by NAV fit rather than cash-flow fit.

9.3 Telescoping and asymptotic nesting

The discounted Euler errors telescope, which establishes that the Euler approach nests the standard cash-flow approach.

Proposition (Telescoping). *For a fund with inception value* $\mathrm{NAV}_{0,i} = 0$ *and terminal value* $\mathrm{NAV}_{T_i,i} = 0$ *(fully liquidated), the sum of discounted Euler errors equals the standard cumulative pricing error:*

$$\sum_{t=1}^{T_i} \Psi_{0,t-1}(\theta)\, \eta_{t,i}(\theta) = \sum_{t=1}^{T_i} \Psi_{0,t}(\theta)\, CF_{t,i}.$$

Proof. Expand the left-hand side using $\Psi_{0,t-1} \cdot \Psi_{t-1,t} = \Psi_{0,t}$:

$$\begin{aligned} \sum_{t=1}^{T_i} \Psi_{0,t-1}\, \eta_{t,i} &= \sum_{t=1}^{T_i} \left[\Psi_{0,t}\, (CF_{t,i} + \mathrm{NAV}_{t,i}) - \Psi_{0,t-1}\, \mathrm{NAV}_{t-1,i} \right] \\ &= \sum_{t=1}^{T_i} \Psi_{0,t}\, CF_{t,i} \;+\; \underbrace{\sum_{t=1}^{T_i} \left[\Psi_{0,t}\, \mathrm{NAV}_{t,i} - \Psi_{0,t-1}\, \mathrm{NAV}_{t-1,i} \right]}_{\text{telescopes to } \Psi_{0,T_i} \mathrm{NAV}_{T_i} - \mathrm{NAV}_{0,i} \,=\, 0}. \end{aligned}$$

□

This identity has three consequences:

1. **Nesting.** For fully liquidated funds, the Euler pricing error equals the standard cash-flow pricing error. If all vintages are seasoned, the estimator reduces to a pure cash-flow estimator.
2. **Convergence.** As an unseasoned fund matures and its NAV declines toward zero, the Euler error converges continuously to the cash-flow error. There is no discontinuity at the seasoned/unseasoned boundary.
3. **Asymptotic irrelevance.** Only the most recent vintages within a bounded horizon D carry nonzero NAV terms. As $V \to \infty$, the fraction $|\mathcal{V}^U|/V \to 0$. The estimator is therefore *asymptotically a pure cash-flow estimator*; the Euler restrictions are finite-sample corrections that vanish in the limit. To make the argument concrete: in a mature dataset spanning 1,000 vintage years, perhaps 15 recent vintages would be classified as unseasoned under any reasonable threshold δ. The remaining 985 vintages contribute pure cash-flow pricing errors with no NAV dependence whatsoever. The NAV-based Euler restrictions therefore constitute roughly 1.5% of the moment conditions by vintage count. As the panel grows, this fraction shrinks monotonically toward zero.

Remark: NAV quality and the Euler moment condition. The telescoping identity is a pure algebraic result that holds for *any* sequence of reported NAVs—appraisals, mark-to-model values, or even arbitrary numbers. It does not depend on NAVs being economically "correct." However, the population Euler moment condition $E[\eta_{t,i}(\theta_0)] = 0$ makes a stronger claim: that the beginning-of-period NAV equals the risk-adjusted present value of within-period cash flows plus the end-of-period NAV. For traded assets, no-arbitrage forces this condition to hold. For private-equity funds, NAVs are appraisals whose co-movement with public factors may be smoothed or biased, so the Euler restriction holds only approximately.

The paper's design controls this concern in three ways. First, for fully liquidated funds, the telescoping identity ensures that all intermediate NAV terms cancel exactly—NAV quality is irrelevant for seasoned vintages, regardless of how inaccurate the intermediate appraisals were. Second, the asymptotic irrelevance property ensures that the fraction of moment conditions relying on NAV quality vanishes as the panel grows. Third, even for unseasoned vintages in finite samples, the one-period Euler restriction is more robust to NAV measurement error than the alternative of discounting a large terminal NAV over the fund's entire remaining horizon: appraisal error in a one-quarter NAV change is a much smaller concern than a systematic valuation bias compounded over many years.

9.4 Why one-period restrictions stabilize alpha

Under the standard cumulative approach, including an unseasoned fund with residual $\text{NAV}_T > 0$ requires discounting by the cumulative SDF $\Psi_{0,T}(\theta)$, which contains the factor $(1+\alpha)^{-T}$. For large T or large NAV_T, a small perturbation in α produces a large swing in the pricing error. This mechanism could aggravate the exploding-alpha issue.

The quarterly Euler restriction replaces this single long-horizon discounting with T one-period discountings, each involving $(1+\alpha)^{-1}$. The sensitivity to α is proportional to the quarter length Δt, not exponential in the fund horizon T. Moreover, each quarterly error $\eta_{t,i}$ subtracts NAV_{t-1} from a discounted quantity of comparable magnitude, so the error is well-conditioned regardless of fund size.

9.5 Integration into fold moments

The fold moment retains its original form:

$$m_{n,r}(\theta; M, \eta) = \sum_{v \in \mathcal{V}^S \cap A_{n,r}} \omega_{n,v,r}\, \epsilon^S_{n,v,r}(\theta; M, \eta) \;+\; \sum_{v \in \mathcal{V}^U \cap A_{n,r}} \omega_{n,v,r}\, \epsilon^U_{n,v,r}(\theta; M, \eta),$$

where ϵ^S is the standard cumulative cash-flow pricing error and ϵ^U is the Euler-based error defined above. Both terms are pricing errors that equal zero under correct pricing. The entire estimation and validation machinery from Sections 5 and 6 applies without modification—only the construction of vintage-level pricing errors differs between seasoned and unseasoned vintages.

10 Relation to DLP12 and KN16

The paper can be positioned in one paragraph:

- **From DLP12:** price private cash flows directly with an SDF.
- **From KN16:** use aggregated pricing moments and an NPV-style abnormal-performance object.
- **New here:** treat the construction of PE pricing moments as a resampling problem and rank SDF models by holdout pricing errors across many admissible fund assignments.
- **Also new:** Euler NAV restrictions for unseasoned vintages that telescope into the standard cash-flow restriction, avoiding both sample waste and exploding-alpha degeneracy (Section 9).

The paper is therefore not a repair of the existing SDF paper. It is a new empirical-design paper: how should we build, resample, and validate PE pricing moments when the sample is small and the moment construction is consequential?

11 Empirical Algorithm

For each candidate model M, alpha design η, and partition scheme Π:

1. Partition vintages into seasoned ($\mathcal{V}^S$) and unseasoned ($\mathcal{V}^U$) sets (Section 9).
2. Choose P estimation folds and h holdout folds.
3. Draw a sampled assignment n from Π.
4. Build fold cash-flow portfolios and fold pricing moments $m_{n,r}(\theta; M, \eta)$.
5. Estimate $\hat{\theta}_n(M, \eta, \Pi)$ on the P estimation folds.
6. Evaluate the fitted model on the holdout fold.
7. Repeat over many sampled assignments.
8. Rank model-design pairs by average holdout pricing loss.

Alongside the mean holdout loss, report diagnostics that help interpret the ranking:

- median and dispersion of holdout losses,
- dispersion of $\hat{\theta}_n$ across assignments,
- frequency of non-convergence,
- frequency of boundary or exploding-alpha solutions,
- sensitivity of results to within-vintage resampling versus shuffled vintage-year portfolios.

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